MATTIA PIAZZA

Researcher
Mechatronics Engineer

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WHO AM I?

I am a Mechatronics Engineer and doctoral researcher specializing in motion planning, optimal control, and real-time optimization for autonomous systems and robotics. My work bridges theoretical algorithm development and practical deployment in complex, dynamic environments (including autonomous racing and connected mobility). With strong expertise in C++, Python, and software engineering, I develop high-fidelity simulators and sampling-based motion planning algorithms. I am eager to apply my expertise in continuous motion planning to broader automated planning and scheduling challenges in robotics. I strive for excellence in both engineering and personal growth, bringing curiosity, ambition, and a focused, determined mindset to every challenge I take on.

SOFTWARE / PROGRAMMING SKILLS

- C/C++
- Matlab
- Python
- Git
- CMake/make
- Maple

KEY QUALIFICATIONS

PhD in Systems Engineering	Research on optimal control and lap-time simulation for racing applications.
4+ years experience	Motion planning, Optimal Control, and Vehicle and Autonomous Agents dynamics.
Vehicle Dynamics	Developed custom multi-body models and controls for OEMs and motorsport.
Programming Proficiency	Extensive use of C++, MATLAB and Python for algorithm and tool development.
Software Engineering	Experienced with Git, and cross-platform development workflows.
International Collaboration	Experience in European, National and OEM founded projects.

WORKING EXPERIENCES

2022–Present **PhD Researcher - Doctoral program in Mechatronics and Systems Engineering*** — **University of Trento, IT**
Motion Planning Algorithms for autonomous driving, multiagent cooperation and racing scenarios.
Multiplatform real-time control library. MLT simulator for MotoGP with Indirect Optimal Control Methods.

2025–Present **Visiting Researcher - Autonomous Vehicle Systems Laboratory** * — **Technische Universität München (TUM), DE**
Velocity profile online optimization for Indy Autonomous Challenge (IAC) and Abu Dhabi Autonomous Racing League (A2RL). Time-optimal velocity planning and control under dynamic grip constraints.

2021–2022 **Research Fellowship (Research grant decree n. 205/2020)** — **University of Trento, IT**
Solution algorithms comparison for minimum-time optimal control problems for racing vehicles.

2020–2021 **Research scholarship (Decree n. 74/2020)** — **University of Trento, IT**
Minimum Time Manoeuvres of complex vehicle models described with DAEs.

* Employer references available upon request or provided as academic/project endorsements.

PROJECTS

2022–2024 **VeDi 2025 -** *Ministero Italiano dello Sviluppo Economico* — **CRF - STELLANTIS | University of Trento**
Development of a motion planner for maneuver negotiation between AVs via V2X communication.

2024–2025 **SUNRISE** **(Safety assUraNce fRamework for connected, automated mobIlity SystEms)** — **European Project | University of Trento**
Horizon Europe project - Safety Assurance Framework for CCAM.

2025–Present **LUIGI** **(Lower Upper Iterative Gggv Integration for minimum lap time)** — **TUM IAC and A2RL racing team | University of Trento**
Developed a fast real time velocity profile optimization tool under generic complex gggv constraints.

2021–Present **Minimum Lap Time Simulator** — **APRILIA RACING | University of Trento**
With Aprilia Racing - Developed an Optimal Control offline MLT simulator and performance analysis.

LANGUAGES

	Listening	Reading	Spoken Interaction	Spoken Production	Writing
Italian (mother tongue)	C2	C2	C2	C2	C2
English	C1	C1	C1	C1	C1
German	A1	A2	A1	A1	A1

EDUCATION

2022–Present **PhD Candidate - Doctoral program in Materials, Mechatronics and Systems Engineering** **University of Trento**

2020 **Professional qualification to practice as an engineer - Industrial Engineer** **University of Trento**

2017–2020 **Master of Science degree in Mechatronics Engineering - Mechanics e Mechatronics** **University of Trento**

2013–2017 **Bachelor of Science degree in Mechanical Engineering** **University of Udine**

TEACHING EXPERIENCES

2023–2026 **Teaching assistant** **University of Trento**
Courses: *Mechatronics Systems Simulation*, *Sistemi Meccanici e Modelli* and *Fondamenti di Meccanica*
Prof. F. Biral, Prof. E. Bertolazzi. and Prof. M. Da Lio.

2018–2019 **Tutor specific areas for mechanics and mechatronics and video-lectures** **University of Trento**
Frontal lecturer and video lecture (DII Youtube channel).

PUBLICATIONS

1. **Mattia Piazza**, Mattia Piccinini, Sebastiano Taddei and Francesco Biral. “MPTree: A Sampling-based Vehicle Motion Planner for Real-time Obstacle Avoidance”. In: *IFAC-PapersOnLine* 58.10 (2024), pp. 146–153. DOI: `10.1016/j.ifacol.2024.07.332`
2. Antonello Cherubini, Gastone Pietro Rosati Papini, Alice Plebe, **Mattia Piazza** and Mauro Da Lio. “Bootstrapped Neural Models for Predicting Self-Driving Vehicle Collisions With Quantified Confidence: Offline and Online Applications”. In: *IEEE Transactions on Intelligent Vehicles* (2024). DOI: `10.1109/TIV.2024.3512786`
3. **Mattia Piazza**, Mattia Piccinini, Sebastiano Taddei, Francesco Biral and Enrico Bertolazzi. “Real-Time Velocity Profile Optimization for Time-Optimal Maneuvering With Generic Acceleration Constraints”. In: *IEEE Robotics and Automation Letters* 11.2 (2026), pp. 1674–1681. DOI: `10.1109/LRA.2025.3643297`
4. **Mattia Piazza**, Enrico Bertolazzi and Marco Frego. “A Non-Smooth Numerical Optimization Approach to the Three-Point Dubins Problem (3PDP)”. in: *Algorithms* 17.8 (2024), p. 350. DOI: `10.3390/a17080350`
5. Mattia Piccinini, Sebastiano Taddei, Matteo Larcher, **Mattia Piazza** and Francesco Biral. “A Physics-Driven Artificial Agent for Online Time-Optimal Vehicle Motion Planning and Control”. In: *IEEE Access* 11 (2023), pp. 46344–46372. DOI: `10.1109/ACCESS.2023.3274836`
6. Mattia Piccinini, Sebastiano Taddei, **Mattia Piazza** and Francesco Biral. “Impacts of ggv Constraints Formulations on Online Minimum-Time Vehicle Trajectory Planning”. In: *IFAC-PapersOnLine* 58.10 (2024), pp. 87–93. DOI: `10.1016/j.ifacol.2024.07.323`
7. **Mattia Piazza**, Alexander Langmann, Francesco Biral, Johanness Betz and Mattia Piccinini. “Beyond the Apex: Online Minimum-time Velocity Planning on Three-Dimensional Paths with Variable Friction”. In: *IEEE RA-L (Under review)* (2026)
8. Mattia Piccinini, Patrick Zambiasi, Aniello Mungiello, **Mattia Piazza**, Felix Jahncke and Johannes Betz. “Trajectory Planning and Control near the Limits: an Open Experimental Benchmark on the RoboRacer Platform”. In: *IEEE ITSC conference (Accepted)*. 2026
9. Giacomo Corradini, **Mattia Piazza**, Davide Stocco and Francesco Biral. “Corner Classification for Driver Style Identification”. In: *AVEC 2026 conference (Accepted)*. 2026
10. **Mattia Piazza**, Filippo Faccini, Gioele Defrancesco, Giacomo Corradini, Gastone Pietro Rosati Papini and Francesco Biral. “Fast GGV-Compliant Sampling-Based Planning for Vehicles”. In: *AVEC 2026 conference (Accepted)*. 2026

AWARDS

Jul 2024 **IFAC Young Author Award** **IFAC CTS 2024 | Ayia Napa, Cyprus**
IFAC Young Author Award at the 17th Symposium on Control of Transportation Systems for the conference paper titled: "MPTree: A Sampling-based Vehicle Motion Planner for Real-time Obstacle Avoidance"

Nov 2023 **Best Poster Award** **Department of Industrial Engineering (DII), University of Trento | Trento, Italy**
First place award for best poster award for research titled "MPTree: Motion primitive tree exploration for trajectory planning with dynamic and cooperative obstacle avoidance"

HOBBIES AND SPORTS

Outside of work, I am passionate about both technology and the outdoors. I enjoy climbing, skiing, hiking, trail running, and ski touring—activities that challenge me physically and mentally. As an amateur climber and alpinist, I find joy in pushing my limits and exploring new terrain. Time in the mountains helps me disconnect from technology and reconnect with nature, bringing balance and perspective to my life.

PRIVACY

In compliance with the Italian Legislative Decree no. 196 dated 30/06/2003, I hereby authorize the recipient of this document to use and process my personal details for the purpose of recruiting and selecting staff and I confirm to be informed of my rights in accordance to art. 7 of the above-mentioned decree.

Trento, 19th May 2026

Mattia Piazza